

# The AI ladder: retrieval, adapters, humans, and who holds the keys

Knowledge goes in retrieval, behaviour goes in adapters, what neither can answer goes to a person — and no model, at any rung, decides what it is allowed to do.

The one-line version: knowledge goes in retrieval, behaviour goes in adapters, what neither

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can answer goes to a person, and no model — at any rung — decides what it is allowed to do. That decision lives in one place, before every action, and an agent reaches it only through a mandate.

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## 0. Vocabulary

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Four readers use these words — a business analyst, a developer, a compliance reviewer and a designer — and several of them are used loosely elsewhere. This document uses them this way.

| TERM                                                   | MEANS HERE                                                                                                                                                                                                                               | DOES NOT MEAN                                                                                                                         |
|--------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>RAG</b> — retrieval-augmented generation            | The model answers from passages retrieved at question time (here: a vector index of FAQs, articles, manuals)                                                                                                                             | Training. Nothing about the model changes                                                                                             |
| <b>CRAG</b> — <i>Corrective</i> RAG (Yan et al., 2024) | RAG with a <b>grader between retrieval and answer</b> : passages judged correct → used; incorrect → discarded and another source tried; ambiguous → both                                                                                 | A different retriever. It is a gate on whatever the retriever returns                                                                 |
| <i>CRAG (benchmark)</i>                                | Meta's Comprehensive RAG Benchmark (KDD Cup 2024) — a test set                                                                                                                                                                           | The technique above. Check which one is meant                                                                                         |
| <b>LoRA</b> — Low-Rank Adaptation (Hu et al., 2021)    | Base model frozen; small low-rank matrices trained beside chosen layers (well under 1 % of parameters); shipped as a swappable <b>adapter</b> of megabytes. <b>QLoRA</b> is the same on a compressed base                                | A full fine-tune, or a way to add facts                                                                                               |
| <b>Human-in-the-loop escalation</b>                    | A question the system cannot answer well enough is routed to a person, on a clock set by its urgency, and the person's answer is kept                                                                                                    | A "contact us" link. An escalation has a tier, a deadline and a record                                                                |
| <b>Permission</b>                                      | What an <i>application</i> may reach                                                                                                                                                                                                     | What a person may do                                                                                                                  |
| <b>Entitlement</b>                                     | What a <i>subject</i> may do (a capability row, revocable)                                                                                                                                                                               | A role name                                                                                                                           |
| <b>Mandate</b>                                         | What an <i>agent</i> may do <b>on someone's behalf</b> : signed, capped, scoped, time-boxed, revocable                                                                                                                                   | A stored password or API key                                                                                                          |
| <b>ADK</b> — Google Agent Development Kit              | A runtime for LLM <b>agents</b> : sub-agents, tools, sessions, memory                                                                                                                                                                    | Infrastructure. It decides what to <i>try</i> , not what is <i>allowed</i>                                                            |
| <b>Kubernetes / Helm</b>                               | Kubernetes runs containers; <b>Helm</b> packages them: a <i>chart</i> (templates) + a <i>values</i> file per deployment                                                                                                                  | A security boundary for data. It places workloads                                                                                     |
| <b>Helmet</b>                                          | This estate's name for a <b>higher-order function</b> at the Cloudflare edge: it takes the page's behaviour (or an agent's tool call) and returns a wrapped version that loads consent first and checks the mandate before anything acts | Helm. The name is borrowed on purpose — chart ≈ loader, values ≈ per-tenant config — but it runs on every request, not once at deploy |

# 1. The challenge

What goes wrong, described by what it looks like from outside:

| FAILURE                            | WHAT THE CUSTOMER SEES                                            | WHY IT HAPPENS                                                           |
|------------------------------------|-------------------------------------------------------------------|--------------------------------------------------------------------------|
| Confident wrong answer             | A fluent reply that cites the wrong product or an old policy      | Plain RAG answers from whatever it retrieved, relevant or not            |
| Stale fact in the weights          | Last month's price, stated as fact, with no source to check       | A fact was trained into a model, where it cannot be corrected or cited   |
| The question nobody owns           | "Sorry, I can't help with that" — and nothing happens next        | No path from a failed answer to a person, and no record that it failed   |
| An agent with standing credentials | A purchase nobody approved, or one over budget                    | The agent held a key instead of asking for permission per action         |
| Two languages, one answer          | The Korean page is right, the English one is not (or the reverse) | Answers maintained per language drift apart                              |
| Configuration that is really code  | A new market deploys and asks the wrong consent questions         | Where data lives was a constant in code, not a setting of the deployment |

## 2. The solution: climb only when the data says so

| RUNG                                   | CHANGES                                                         | USE IT FOR                                                                                                | CLIMB WHEN...                                                                                          | WHAT IT GIVES UP                                              |
|----------------------------------------|-----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| 1.<br><b>Instructions<br/>+ claims</b> | The prompt and the subject's claims (language, market, consent) | Tone, rules, language                                                                                     | Answers lack <b>facts</b> the model cannot know                                                        | Nothing — this is the floor                                   |
| 2. RAG                                 | What the model <b>reads</b>                                     | <b>Facts that change:</b> prices, stock, policy, FAQs, specs, legal text — per tenant, bilingual, citable | Retrieval returns the <b>right</b> passages but answers are still wrong in <b>how</b> they are written | A search index to maintain                                    |
| 2b. CRAG gate                          | Whether retrieved passages are <b>trusted</b>                   | Refusing a weak match, trying the next source, or escalating                                              | — (it belongs on every RAG path)                                                                       | Some questions are answered "I don't know" instead of guessed |
| 3. LoRA adapter                        | How the model <b>writes</b>                                     | Behaviour: tone, domain vocabulary, a fixed format, classifying a question                                | One adapter cannot hold the behaviours needed                                                          | Training data, an evaluation set, an adapter per tenant       |
| 4. Full fine-tune                      | The whole model                                                 | Rarely justified in commerce                                                                              | —                                                                                                      | Cost, and every fact baked in goes stale                      |

**The rule that does not bend:** anything that changes stays at rung 2. A price trained into weights is wrong the day it changes and cannot be cited when a customer disputes it.

**When is the data ready for rung 3?** Not by volume — by measurement:

1. **An evaluation set first.** Real questions with accepted answers, in each language served. Without one, "the model got worse" and "the model got better" are both guesses.

- 2. **The evaluation shows retrieval is right and the writing is wrong.** That is the only signal a LoRA adapter can fix.
- 3. **Hundreds to low thousands of clean question → accepted-answer pairs** for *one* behaviour. Consistency beats size. (A rule of thumb, not a measured threshold.)

### 3. CRAG as it is actually built

The corrective gate is not new here; it is three thresholds that already refuse weak answers.

| SURFACE                               | GRADER                              | BELOW THE BAR                                                                      |
|---------------------------------------|-------------------------------------|------------------------------------------------------------------------------------|
| Commerce FAQ answer                   | Weighted FAQ match score $\geq 0.5$ | Falls through to the next source (web search → vector index of articles → manuals) |
| Ask-the-Docs                          | Vector similarity $\geq 0.45$       | Answers "the documentation does not mention this" – never a guess                  |
| Bilingual product FAQ (floor heating) | Retrieval score $\geq 0.5$          | Escalates to a person (§4)                                                         |

**What it costs:** a strict gate says "I don't know" more often. That is the correct trade in commerce, and it is also the input the next section depends on.

### 4. Human-in-the-loop escalation

A refused answer is not an ending. It becomes an escalation with a tier, a clock and a record.

| TIER   | MEANING                                    | WHO ANSWERS, HOW FAST                                                                                                     |
|--------|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| none   | Answered from the knowledge base           | —                                                                                                                         |
| tier 1 | Must be answerable <b>without</b> a person | Automated — because the Korean team's office hours are 19:00–04:00 US Eastern, a US-morning question cannot wait for them |
| tier 2 | Needs a person, not urgent                 | Next business window                                                                                                      |
| tier 3 | Safety or legal                            | Immediately. A product-safety report is a compliance event: the US CPSC §15(b) clock runs <b>24 hours</b>                 |

## Design rules that make escalation useful, not just polite:

| RULE                                                           | WHY                                                                                                                                     |
|----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| The escalation record accepts no name, email or phone          | A question is training data; a person is a data subject. Mixing them gives the training set a retention clock and a deletion obligation |
| An answer counts as done only when it exists in both languages | Derived by the server ( <code>pair_status</code> ), never asserted by the caller — the invariant the log exists to protect              |
| Keep what the knowledge base returned, even when wrong         | The wrong retrievals are how gaps are found                                                                                             |
| A safety report can arrive through any form                    | Someone whose floor is smoking types it into whatever box is in front of them; the form path flags it too                               |

**Why this feeds the ladder:** every escalation is a question the knowledge base could not answer, paired with the answer a person gave — in both languages. That is the curated, personal-data-free dataset rung 3 needs. The humans are not a fallback for the AI; they are where its next training set comes from.

# 5. ADK vs Kubernetes/Helm vs the Cloudflare helmet

Three layers that are often confused because each "runs" something.

|                         | GOOGLE ADK                                                 | KUBERNETES + HELM                                           | CLOUDFLARE HELMET (HIGHER-ORDER FUNCTION)                                      |
|-------------------------|------------------------------------------------------------|-------------------------------------------------------------|--------------------------------------------------------------------------------|
| What it runs            | An AI agent: which tool to call next                       | Containers: a model server, a gateway, a database           | Every page load and every agent tool call at the edge                          |
| When it acts            | During a conversation                                      | At deploy time (Helm renders once)                          | On every request                                                               |
| Its "config"            | Agent definitions, session state                           | A chart + a values file per deployment                      | A loader + per-tenant config ( /stack/config ) – the values file of a web page |
| What it decides         | What to try                                                | Where workloads run                                         | Whether an action may proceed – consent first, then the mandate                |
| Rollback                | Redeploy the agent                                         | One command, with release history                           | Only via a worker deploy – per-tenant promote/rollback is not built            |
| Needed by every client? | No – optional (Google-side agents, BigQuery insights, A2A) | Only where a tenant self-hosts (e.g. a database in-country) | Yes – it is where permission is enforced                                       |

**The equivalence that holds:** an endpoint with permission checks in its function stack is an ADK **tool** with a guard – not an ADK **agent**. The agent is the layer that chooses which tools to call. Because the permission lives in the endpoint, the agent is swappable: ADK, an edge model, or a database vendor's own agent can sit on top, and none of them can do more than the endpoint allows.

**Helm and the helmet share a shape, not a job.** Both are *one template, different values per tenant*. Helm places a model server in Seoul or Iowa; the helmet decides, on the request, that a Korean visitor must grant an itemised cross-border consent before anything is stored – and that decision itself depends on

a value (where this deployment keeps personal data), because a Seoul deployment must *not* ask it.

## 6. Mandates: the only way an agent acts

The helmet wraps every agent action in the same check. The agent brings a **mandate**, never a key.

| PROPERTY      | WHAT IT ENFORCES                                                                                         | CHECKED WHERE                                                             |
|---------------|----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| Signed        | Issued by the platform for this subject; forged mandates fail verification                               | Server, and verifiable offline with the public key                        |
| Capped        | A maximum amount — checked on the <b>discounted</b> total, so a discount cannot be used to slip under it | Server, at checkout                                                       |
| Scoped        | Which rails and markets it covers (e.g. a Korean settlement requires a sovereign-settlement capability)  | Server, before any draft order exists                                     |
| Time-boxed    | Expires (default 24 hours)                                                                               | Server                                                                    |
| Revocable     | One row change; takes effect on the next call                                                            | Server — entitlements are resolved per request, never frozen into a token |
| Consent-bound | Requires the subject's agent-purchase consent                                                            | Server                                                                    |

**Why a higher-order function and not a check inside each tool:** a check written into each tool is a check someone can forget in the next tool. A function that *wraps* every tool call — takes the action and returns the guarded action — is written once, and a new tool is guarded by construction. It is the same reason consent loads first on every page rather than per feature.

**What an agent never holds:** a payment card (payment finishes on the processor's hosted page), a standing API key, or the shopper's login beyond the

single step it was handed for.

## 7. What this costs

| CHOICE                           | COST                                                                                      | STATED PLAINLY                                                                                                      |
|----------------------------------|-------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| CRAG gates                       | More "I don't know"                                                                       | Correct in commerce, visible to customers                                                                           |
| Escalation with tiers            | People on a clock, in two languages                                                       | Tier 1 must be automatable because the humans are asleep for half the day                                           |
| LoRA later, not now              | Requires an evaluation set before any training                                            | Slower to "AI-first" demos; faster to trustworthy ones                                                              |
| ADK optional                     | Google-specific features (Agent Engine memory, A2A) only where a tenant uses Google Cloud | Agent Engine sessions and memory are <b>not yet</b> covered by the erasure sweep — demo tenants only until they are |
| Edge models for public questions | Cloudflare Workers AI does not pin inference to one country                               | Fine for catalogue and FAQ; personal-data answers for Korea need a Korean-hosted model                              |
| The helmet                       | Runs on every request, so a bad config reaches visitors at once                           | Helm's one-command rollback is the property most worth borrowing, and it is not built                               |

# 8. Summary

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| QUESTION                                  | ANSWER                                                                                          |
|-------------------------------------------|-------------------------------------------------------------------------------------------------|
| Where do facts go?                        | Retrieval (rung 2), behind a corrective gate                                                    |
| Where does behaviour go?                  | An adapter (rung 3) – only after an evaluation shows the need                                   |
| What happens when neither is good enough? | A tiered escalation to a person, whose answer becomes the next training pair                    |
| Is ADK required?                          | No. It is one possible agent layer; permission does not live in it                              |
| What do Kubernetes/Helm do here?          | Place workloads per deployment; the values file is where residency is declared                  |
| What stops an agent overreaching?         | A signed, capped, scoped, time-boxed, revocable mandate – checked by the helmet on every action |